JAN–I–631

10 AUGUST 1948

NATIONAL MILITARY ESTABLISHMENT SPECIFICATION

INSULATION, ELECTRICAL, SYNTHETIC-RESIN COMPOSITION, NONRIGID

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Navy Department
17-I-17
1 Sept. 1945

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATION

A–1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATIONS

JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.
JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.
JAN–B–107—Boxes, Wood, Wirebound (Overseas Type).[1]
JAN–P–108—Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
JAN–P–125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.
JAN–P–127—Packaging and Packing for Overseas Shipment—Tape, Adhesive, Pressure-Sensitive, Water Resistant.
JAN–P–139—Packaging and Packing for Overseas Shipment—Plywood, Container Grade.

U. S. ARMY SPECIFICATIONS

94–40645—Marking; Exterior, Domestic and Export Shipment, by Contractors.[2]
100–2—Standard Specification for Marking Shipments by Contractors.[3]

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material,[4] and Appendix I—Rubber and Synthetic Rubber; General Specification (Methods of Tests).

U. S. AIR FORCE SPECIFICATION

40985—Marking of Interior Packages (for Communications Equipment).[2]

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[1] Not applicable to Air Force purchases.
[2] Applicable only to Air Force purchases.
[3] Applicable only to Army purchases.
[4] Applicable only to Navy purchases.

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FEDERAL SPECIFICATION

VV-O-401—Oil, Insulating (for Transformers, Switches, and Circuit Breakers).

A-2. *Other publication.*—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[5]

B. CLASSIFICATION

B-1. *Types and grades.*—Electrical insulation covered by this specification shall be of the following types and grades, as specified in the contract or order (see pars. H-2b and H-7):

Type A—Polyethylene. (See par. H-1a.)
Type B—Cellulose acetate butyrate. (See par. H-1b.)
Type C—Cellulose acetate. (See par. H-1c.)
Type D—Ethyl cellulose. (See par. H-1d.)
Type E—Vinylidene chloride. (See par. H-1e.)
Type F—Polyvinyl chloride and copolymers of vinyl chloride and vinyl acetate.
Grade a—General purpose. (See par. H-1f.)
Grade b—Low temperature. (See par. H-1g.)

B-2. *Forms.*—Electrical insulation shall be furnished in the following forms, as specified in the contract or order (see par. H-2c):

Form F—Film, in rolls.
Form S—Sheet, flat or in rolls.
Form T—Tape.
Form U—Tubing.

C. MATERIAL AND WORKMANSHIP

C-1. *Material.*—

C-1a. The material shall be of a suitable synthetic-resin composition properly formulated with plasticizers, pigment, fillers, or other ingredients as may be necessary to meet the requirements of this specification, and to provide a high degree of flame resistance and physical resiliency by selection of a flame retardant and plasticizer that are not fugitive in character, and by their incorporation in such a manner that they will be maintained with age and within the temperature limits applicable to the particular type of material.

C-2. *Workmanship.*—The material shall be manufactured and processed in a careful and workmanlike manner in accordance with good practice.

D. GENERAL REQUIREMENTS

D-1. *Qualification approval.*—Electrical insulation supplied under this specification, either directly to the Government or for ultimate sale to the Government, shall be of a brand and type having qualification approval satisfactory to the bureau or service concerned as determined by the tests specified in section F. Contractors proposing to supply equipment using electrical insulation furnished under this specification shall ascertain that the required qualification approval exists. (See par. H-3.)

[5] Applicable only to Navy purchases.

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D–1a. *Changes.*—If the manufacturer desires to change his process or the composition of his material, after qualification approval has been granted, he shall submit to the activity concerned a statement to that effect, describing the proposed change together with evidence to substantiate his claim that such change is suitable. Similar notification and substantiating evidence shall be submitted at any later time if a change again becomes necessary or desirable. At the discretion of the activity concerned, test specimens may be required to prove the suitability of the proposed changes.

D–2. *Accuracy of calculations.*—Unless otherwise specified, observed or calculated values shall be rounded off to the nearest digit in the last right-hand place of figures used in expressing the limiting value. (See table XIV.)

D–3. *Uniformity.*—All material in a lot of the same type and brand of any one manufacturer shall be uniform in texture, color, cross-section, finish, and in the properties specified in table XIV.

D–4. *Surface defects.*—Materials shall be free from blisters, bubbles, wrinkles, or cracks and shall be reasonably free from other small defects such as scratches and dents.

D–5. *Colors.*—Materials shall be furnished in the following colors, as specified in the contract or order (see par. H–2 (d)):

Black	Transparent (clear or slightly tinted)
Blue	
Green	White
Red	Yellow

D–6. *Dimensions (see par. H–2e).*—

D–6a. *Films.*—Films shall be furnished in thicknesses of 0.002 inch or less, of widths greater than 3 inches, and in rolls as specified in the contract or order. If the roll of film is subsequently slit into narrow widths for application by taping, the resultant product is then defined as a "tape."

D–6a (1) *Thickness.*—Unless otherwise specified, the standard thicknesses of form F material and of tape, form T, cut from film, shall be as shown in table I, as specified in the contract or order.

TABLE I.—*Standard thicknesses, forms F and T.*

Thickness	Tolerance (plus or minus)
Inch	*Inch*
0.0005	0.00025
.0010	.00025
.0015	.00025
.0020	.0003

D–6b. *Sheets.*—Sheets shall be furnished in thicknesses greater than 0.002 inch, in widths greater than 3 inches. If the sheet is subsequently slit into narrow widths for application by taping, the resultand product is then defined as "tape." Thin sheets shall be supplied in rolls if so specified in the contract or order.

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D–6b (1). *Thickness.*—Unless otherwise specified, the nominal thicknesses of form S material and of tape, form T, cut from sheets, shall be as follows, as specified in the contract or order:

Inch	*Inch*
0.003	0.020
.004	.025
.005	.030
.008	.040
.010	.050
.015	.060

D–6b (1) a. *Tolerance.*—Permissible variations from standard thicknesses of the various types shall be ±10 percent. At least 90 percent of the area of the sheet shall be within this variation, and at no point shall the thickness as measured vary from the nominal by a value greater than 12.5 percent.

D–6c. *Tape.*—Tape shall be of any specified thickness (see par. H–6), furnished in narrow-width rolls of 3 inches or less, wound over a suitable cylindrical core for subsequent application by hand or machine taping.

D–6c (1). *For application by machine.*—Where the material is required for machine, taping the width and length of material on the roll, the over-all diameter of the roll, and the mandrel diameter of the cylindrical core shall be as specified in the contract or order.

D–6c (2). *For application by hand.*—Where the material is required for application by hand (hand taping) it shall, unless otherwise specified, be put up in rolls approximately 4½ inches in diameter, wound over a suitable 1- or 1½-inch core, convenient for hand application. Unless otherwise specified, tape for hand application shall be furnished in the standard widths shown in table III, as specified in the contract or order.

TABLE III.—*Standard widths, form T.*

Width	Tolerance (plus or minus)
Inches	*Inch*
0.50	0.02
.75	.02
1.00	.02
1.25	.04
1.50	.04
2.00	.04
2.50	.04
3.00	.04

D–6c (3). Tape shall be uniform, even, and smooth in appearance with a surface entirely free from holes. The edges of the tape shall be straight, true, and unbroken. If tapes are manufactured by slitting film or sheet, the method shall be such that the edges are unbroken and tensile strength and elongation do not depreciate more than 15 percent of the value obtained on the original sheet or film. It shall be tightly wound in rolls with a suitable separator, when necessary, between layers, to prevent the material from adhering to itself when unwound from the original roll. When unwound from the roll, the

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tape shall remain reasonably flat with no tendency to curl transversely. The tape, when applied, shall adhere snugly without wrinkling, curling, or buckling.

D–6d. *Tubing.*—Tubing shall be of any specified wall thickness, furnished in tube form in lengths of 25 feet, unless otherwise specified. If in continuous lengths, tubing shall be furnished on reels suitable for mounting on a spindle.

D–6d (1). Unless otherwise specified, tubing shall be furnished in the standard inside diameters and wall thicknesses shown in table IV, as specified in the contract or order.

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TABLE IV.—*Inside diameters and wall thicknesses, form U.*

Size No.	Inside diameter (inches)			Wall thickness (inches)		
	Nominal	Minimum	Maximum	Minimum	Maximum	Nominal
20	0.034	0.032	0.039	0.013	0.019	0.016
19	.038	.036	.044	.013	.019	.016
18	.042	.040	.049	.013	.019	.016
17	.047	.045	.054	.013	.019	.016
16	.053	.051	.061	.013	.019	.016
15	.059	.057	.067	.013	.019	.016
14	.066	.064	.072	.013	.019	.016
13	.076	.072	.080	.013	.019	.016
12	.085	.081	.089	.013	.019	.016
11	.095	.091	.101	.013	.019	.016
10	.106	.102	.112	.013	.019	.016
9	.118	.114	.124	.017	.023	.020
8	.133	.129	.141	.017	.023	.020
7	.148	.144	.158	.017	.023	.020
6	.166	.162	.178	.017	.023	.020
5	.186	.182	.198	.017	.023	.020
4	.208	.204	.224	.017	.023	.020
3	.234	.229	.249	.017	.023	.020
2	.263	.258	.275	.017	.023	.020
1	.294	.289	.311	.017	.023	.020
0	.330	.325	.347	.017	.023	.020
5/16	----------	.313	.334	.022	.028	.025
3/8	----------	.375	.399	.022	.028	.025
7/16	----------	.438	.462	.022	.028	.025
1/2	----------	.500	.524	.022	.028	.025
5/8	----------	.625	.655	.025	.035	.030
3/4	----------	.750	.786	.030	.040	.035
For cable size (cir. mils)	For cable end sealing [1]					
3,000	0.160	0.155	0.170	0.018	0.023	0.020
4,000	.250	.245	.260	.018	.023	.020
9,000	.280	.275	.290	.018	.023	.020
14,000	.320	.315	.330	.018	.023	.020
23,000	.360	.355	.375	.018	.023	.020
30,000	.400	.395	.420	.028	.033	.030
40,000	.430	.425	.450	.028	.033	.030
50,000	.450	.445	.470	.028	.033	.030
60,000	.520	.515	.540	.028	.033	.030
75,000	.570	.565	.590	.037	.043	.040
100,000	.630	.625	.650	.037	.043	.040
125,000	.680	.675	.700	.037	.043	.040
150,000	.760	.750	.785	.037	.043	.040
200,000	.820	.810	.845	.046	.054	.050
250,000	.890	.880	.925	.046	.054	.050
300,000	.960	.950	1.000	.046	.054	.050
350,000	1.030	1.020	1.070	.046	.054	.050
400,000	1.100	1.090	1.150	.046	.054	.050
500,000	1.165	1.150	1.200	.055	.065	.060
650,000	1.225	1.205	1.255	.055	.065	.060
[2] 400,000	1.285	1.260	1.310	.055	.065	.060
[3] 800,000	1.365	1.345	1.400	.055	.065	.060
[2] 650,000	1.470	1.455	1.510	.055	.065	.060
[3] 800,000	1.585	1.565	1.615	.055	.065	.060

[1] Tubing is intended for slipping over the individual conductor insulation for multiple conductor cables and for single conductor cables after the impervious cable sheath has been removed, at the location of the seal, except as specified in footnotes 2 and 3.

[2] For single conductor cable, where tubing is applied over the impervious sheath.

[3] For single conductor cable, where tubing will slip over the conductor insulation (sheath removed) of 800,000-circular-mil cable, and may also be used over the impervious sheath of 500,000-circular-mil cable.

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D–7. *Chemical analysis.*—The results of the chemical analysis made upon material offered under the contract or order shall be approximately the same as that on samples submitted for qualification approval. (See par. F–4o.)

E. DETAIL REQUIREMENTS

E–1. *Qualitative.*—

E–1a. *Effect of heat on materials under tension (forms F, S, and T).*—The material shall not crack or split in layers when tested as described in paragraph F–4g.

E–1b. *Flexibility after aging.*—

E–1b (1). *Oven aging.*—None of the samples of material shall exhibit cracks due to hardening when tested as described in paragraph F–4i (2). In the case of form U, cracking at the creased edges of the bend will be disregarded.

E–1b (2). *Ultraviolet light aging (type F, grade b only).*—Type F, grade b material shall not exhibit cracks due to hardening when tested as described in paragraph F–4i (3). In the case of form U, cracking at the creased edges of the bend will be disregarded.

E–1c. *Cold resistance.*—

E–1c (1). *Cold bend.*—The material shall not crack or break when tested as described in paragraph F–4j (1). In the case of form U, cracking at the creased edges of the bend will be disregarded.

E–1c (2). *Cold vibration (type F, grade b, form U only).*—Type F, grade b, form U material shall not crack or break when tested as described in paragraph F–4j (2).

E–1d. *Oil resistance—flexibility (not applicable to type A and type D).*—

E–1d (1). *Forms F, S, and T.*—The material shall not increase in dimension (width or thickness) by more than 15 percent when tested as described in paragraph F–4l. It shall not break or crack when creased on itself.

E–1d (2). *Form U.*—At the completion of the test described in paragraph F–4l, oil shall not have penetrated into the tubing and the tubing shall not have developed cracks nor increased in diameter at any point by more than 10 percent.

E–1e. *Corrosive effect.*—Copper shall not be corroded or otherwise adversely affected by contact with this material. (See par. F–4m.)

E–1f. *Solvent and acid resistance (type F, grade b only).*—This material shall be resistant both chemically and physically to special testing fuel SR–6, as defined in detail in Appendix I of Navy Department General Specifications for Inspection of Material, to ethylene glycol, and to battery electrolyte (38-percent sulfuric acid by volume). (See par. F–4n.)

E–1g. *Residual stress.*—Except where otherwise specified or approved, the material shall be free from residual stress (internal), i. e., annealed, in the form supplied.

E–1h. *Luminescence (type F, grade b only).*—Type F, grade b material shall not luminesce when tested as described in paragraph F–4p.

E–2. *Quantitative.*—

E–2a. *Property values.*—When specimens are subjected to the tests specified in table XIV, the values obtained from any set of specimens for any specific property after any one conditioning procedure specified in the table shall be averaged, and this average value shall be not

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less than the minimum nor more than the maximum value (as applicable) specified in table XIV.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Qualification tests.*—

F–1a. *Purpose.*—Qualification tests are those tests made at a Government laboratory upon material submitted for qualification approval, in the form of sheet, film, tape, or tubing.

F–1b. *Location.*—Unless otherwise specifically required and approved, qualification tests shall be conducted at a Government laboratory designated by the bureau or service concerned, and unless otherwise specified, will be at the Government's expense.

F–1c. *Type and form approval.*—Separate qualification approval will be required for each type and form of material covered by this specification.

F–1c (1). *Tape cut from film or sheet.*—Where qualification approval has been obtained for a particular brand of film or sheet, it will not be required that complete qualification tests be conducted on tape formed from such film or sheet by slitting, in order to obtain approval for the tape. In such cases it will be necessary to conduct only such tests as are considered necessary to determine that the properties of the material have not been adversely affected by the slitting. Unless otherwise specified, it will be necessary, in order to obtain approval for tape cut from film or sheet, to submit a sufficient number of films or sheets of the same material and thickness for dissipation factor, dielectric constant, and volume resistivity tests. (See par. F–1e.) In the case of tapes formed directly by extrusion, complete qualification tests will be required.

F–1d. *Tests.*—Qualification tests shall be made to determine conformance with the general requirements (see pars. D–3, D–4, and D–5) and with the detail requirements, both qualitative (see par. E–1) and quantitative (see par. E-2) of this specification, and shall consist of the tests listed in table V.

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TABLE V.—*Qualification tests.*

Test	Form		Minimum number of tests for each condition
	F, S, and T	U	
	Paragraph	*Paragraph*	
General inspection	F-4a	F-4a	10
Electrical properties:			
Dielectric break-down	F-4b	F-4b	10
Dielectric constant (forms F, S, and U)	F-4c	F-4c	3
Dissipation factor (forms F, S, and U)	F-4c	F-4c	3
Volume resistivity (forms F, S, and U)	F-4d	F-4d	3
Mechanical properties:			
Tensile strength	F-4e (1)	F-4e (1)	10
Elongation	F-4e (1)	F-4e (1)	10
Percent return (type F, grades a and b)	F-4e (2)	F-4e (2)	3
Bursting strength (type F, grades a and b)		F-4f	3
Thermal properties:			
Effect of heat on material under tension	F-4g		2
Softening temperature (penetration)	F-4h	F-4h	5
Flexibility after oven aging	F-4i (2)	F-4i (2)	3
Flexibility after ultraviolet light aging (type F, grade b only)	F-4i (3)	F-4i (3)	4
Cold bend	F-4j (1)	F-4j (1)	2
Cold vibration (type F, grade b only)		F-4j (2)	2
Lengthwise shrinkage	F-4k	F-4k	3
Miscellaneous:			
Oil resistance-flexibility (not applicable to types A and D)	F-4l	F-4l	3
Corrosive effect	F-4m	F-4m	2
Solvent and acid resistance (type F, grade b only)	F-4n	F-4n	3
Chemical analysis	F-4o	F-4o	1
Luminescense (type F, grade b only)	F-4p	F-4p	1

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F–1e. *Sizes and quantities of material for qualification tests.*—Unless otherwise specified, qualification tests shall be conducted on the sizes of materials shown in table VI. The manufacturer shall furnish the amount of each size as shown in table VI.

TABLE VI.—*Sizes and quantities of material for qualification tests.*

Form	Sizes			Amount of each size
	Size No.	Thickness	Width or i. d. (minimum)	
		Inch	*Inches*	
F	----	0.0005	----	10 square yards of each thickness.
	----	.001	----	
	----	.002	----	
S	----	.005	----	Do.
	----	.010	----	
	----	.020	----	
T [1]	----	.0005	1	1,000 feet of each thickness.
	----	.001	1	
	----	.002	1	
	----	.005	1	
	----	.010	1	
	----	.020	1	
U	18	----	0.040	200 feet.
	13	----	.072	Do.
	6	----	.162	Do.
	----	----	1.02	100 feet.

[1] One square yard of sheet or film material in each thickness will also be required when tape is obtained by slitting. (See par. F–1c (1).) The material shall be furnished in any convenient width over 3½ inches.

F–1f. *Results of tests.*—Qualification approval will be granted when all the sizes of the material as specified in paragraph F–1e have been examined and tested as required herein and have been found to meet the requirements of this specification.

F–2. *Inspection and tests.*—The insulation shall be subject to inspection by authorized Government inspectors. When inspection is conducted at the contractor's plant, tests shall be conducted by the contractor, under the supervision of an authorized Government inspector.

F–2a. *Sampling.*—

F–2a (1) *Lot.*—For purposes of sampling, a lot shall consist of all material of the same form, type, and thickness (forms F, S, and T) or inside and outside diameters (form U) offered for delivery at the same time.

F–2a (2). *Sampling procedure for lot acceptance inspection at the place of manufacture.*—The Government inspector shall select from each lot the number of representative samples shown in table VII for inspection. Each sample shall be of sufficient size so that an adequate inspection can be made.

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TABLE VII.—*Sampling procedure for lot acceptance inspection.*

Number of rolls, sheets, coils, reels, or bundles in lot	Number of rolls, sheets, coils, reels, or bundles to be sampled
1 to 25	All
26 to 50	22
51 to 100	27
101 to 400	33
Over 400	36

F–2a (3). *Sampling procedure for lot acceptance tests at Government laboratory.*—The Government inspector shall select from each lot the number of samples shown in table VIII for tests specified in paragraph F–2c (2), except that the lot acceptance tests at a Government laboratory, as specified in paragraph F–2c, will not be required for lots of tape consisting of 100 rolls or less, or lots of tubing consisting of 1,500 feet or less. (See par. F–2d.)

TABLE VIII.—*Sampling procedure for lot acceptance tests.*

Number of rolls, sheets, coils, reels, or bundles in lot	Number of rolls, sheets, coils, reels, or bundles to be sampled
1 to 5	All
6 and over	5

The minimum quantity of each sample shall be as follows:

Forms F and S—— 9 square feet.
Form T—— 1 roll, or a minimum of 40 feet.
Form U—— 25 feet.

F–2b. *Inspection (at place of manufacture).*—

F–2b (1). *Inspection procedure.*—The Government inspector shall subject the samples selected in accordance with paragraph F–2a (2) to a thorough inspection to ascertain that the dimensions of the materials are in accordance with the values specified in paragraph D–6 and that the material conforms to the requirements of paragraphs D–3, D–4, and D–5. If two or more of the samples fail to meet the requirements of this inspection, or if, in one sample there are two or more noncontiguous portions failing to meet the requirements of this inspection, the entire lot shall be rejected.

F–2b (2). *Action in case of rejection.*—A lot that has been rejected in accordance with the provisions of paragraph F–2b (1) may be resubmitted for Government inspection, but only after the manufacturer, subsequent to the first Government inspection, has done 100 percent inspection or testing of the lot for the deficiency noted and has removed all nonconforming material.

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F-2c. *Lot acceptance tests (at a Government laboratory).*—

F-2c (1). *Lot acceptance procedure.*—Lot acceptance tests shall be conducted at the Government laboratory specified by the bureau or service concerned. If one or more of the samples selected in accordance with paragraph F-2a (3) fails one or more of the tests of paragraph F-2c (2), the lot represented shall be rejected.

F-2c (2). *Tests.*—

F-2c (2) a. All of the tests shown in table IX shall be conducted on each of the samples selected in accordance with paragraph F-2a (3). (Group I.)

TABLE IX.—*Lot acceptance tests for all samples (group I).*

Test	Condition prior to test (see par. F-3)	Minimum number of tests per sample	Form	
			F, S, and T	U
Dielectric break-down	C-96/25/0	3	F-4b	F-4b
Tensile strength and elongation.	C/96/25/50	3	F-4e (1)	F-4e (1)
Percent return (type F, grades a and b).	C-96/25/50	1	F-4e (2)	F-4e (2)
Bursting strength (type F, grades a and b), continuous pressure only.	D_s-48/50 (synthetic sea water).	1		F-4f

F-2c (2) b. All of the tests shown in table X shall be conducted on two of the samples selected in accordance with paragraph F-2a (3). (Group II.)

TABLE X.—*Lot acceptance tests for two samples (group II).*

Test	Condition prior to test (see par. F-3)	Minimum number of tests per sample	Form	
			F, S, and T	U
Dissipation factor and dielectric constant at 1 kc. (types A, B, C, D, forms F, S, and U).	C-96/25/0	3	F-4c	F-4c
Volume resistivity (types A, B, C, D, forms F, S, and U).	C-96/25/0	3	F-4d	F-4d
Softening temperature (penetration).	C-96/25/50	10	F-4h	F-4h
Flexibility after oven aging	E-200/100	5	F-4i (2)	F-4i (2)
Cold bend		2	F-4j (1)	F-4j (1)
Lengthwise shrinkage	E-2/100	3	F-4k	F-4k
Chemical analysis		(1)	F-4o	F-4o

[1] One composite sample.

F-2c (3). *Action in case of rejection.*—A lot that has been rejected in accordance with the provisions of paragraph F-2c (1) may be resubmitted for Government inspection, but only after the manufacturer, subsequent to the first Government inspection, has done 100 percent

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inspection or testing of the lot for the deficiency noted and has removed all nonconforming material.

F–2d. *Lot acceptance procedure for small orders.*—In the case of lots of tape consisting of 100 rolls or less, or lots of tubing consisting of 1,500 feet or less, (see par. F–2a (3)), those of the tests of paragraph F–2c (2) which the Government inspector considers necessary to establish essential equivalence with the approved quality shall be conducted under the supervision of the Government inspector at the place of manufacturer. If one or more of the samples selected fail one or more of the tests, the lot represented shall be rejected.

F–3. *Conditioning.*—Materials shall be conditioned before being tested, as required in table XIV.

F–3a. Designations indicating conditioning of test specimens shall be written as follows:

First: A capital letter indicating the general condition of the specimen, i. e., humidity, immersion, and temperature conditioning.

Second: A number indicating in hours the duration of the conditioning.

Third: A number indicating in degrees centigrade the conditioning temperature.

Fourth: A number indicating relative humidity, whenever relative humidity is controlled. (Relative humidity obtained over calcium chloride shall be taken as zero.)

The numbers shall be separated from each other by slant marks, and from the capital letter by a dash.

F–3b. The following letters shall be used to indicate the respective general conditioning:

Condition C.—The specimens shall be conditioned at the time, temperature, and relative humidity specified.

Examples:

C–96/25/0	96 hours, 25° C., 0 percent relative humidity (desiccation over calcium chloride).
C–96/25/50	96 hours, 25° C., 50 percent relative humidity.
C–96/25/96	96 hours, 25° C., 96 percent relative humidity.

Condition D_s.—The specimens shall be immersed in synthetic sea water for the time and at the temperature specified.

Example:

D_s–48/50 (synthetic sea water)	48 hours, 50° C., in synthetic sea water.

Condition E.—The specimens shall be conditioned at the time and temperature specified.

Examples:

E–400/100	400 hours at 100° C.
E–200/100	200 hours, 100° C.
E–1/100	1 hour, 100° C.

F–3c. *Equipment.*—Conditioning chambers and baths shall be such as to maintain the specified temperature ±2° C., and the specified relative humidity ±2 percent.

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F–4. *Tests.*—The methods and procedures shall be as described in paragraphs F–4a through F–4p. Unless otherwise specified, all tests shall be conducted under room conditions (25° C., 50 percent relative humidity). In the case of materials believed to be oriented, mechanical tests shall be conducted on samples cut lengthwise, and on samples cut crosswise for those properties affected by the orientation.

F–4a. *General inspection.*—Electrical insulation shall be given a thorough general examination to ascertain that the materials and workmanship are in strict conformance with the requirements of this specification. Particular care shall be taken to ascertain that dimensions of the material are in accordance with the values specified in paragraph D–6 and that the material conforms to the requirements of paragraphs D–3, D–4, and D–5.

F–4a (1). *Method for measurement of dimensions.*—

F–4a (1) a. *Thickness, forms F, S, and T.*—The thickness shall be determined by the use of a dial-type micrometer. The lower or fixed anvil shall consist of a flat steel plate 2 inches in diameter; the upper or movable member shall consist of a foot or button having a flat steel face ¼ inch in diameter and arranged so as to move with its face parallel to the face of the anvil at all times. The movable member shall be of such weight, and shall be attached to the operating rod of the micrometer in such a manner, that 3 ounces dead weight shall be applied to all thicknesses of samples being measured. The thickness shall be determined as the average of at least 10 readings taken at representative points on the material. If the thickness so determined falls outside of the specified tolerances, 10 additional readings shall be taken from another section of the material and averaged with the previous readings. The maximum, minimum, and average values shall be recorded.

F–4a (1) b. *Width, form T.*—The width shall be determined by the use of a standard steel scale graduated to read within 0.01 inch. The tape shall be flat on a smooth surface at the time of measurement. At least 10 measurements shall be made, and the maximum, minimum, and average values shall be recorded.

F–4a (1) c. *Diameter and wall thickness, form U.*—A plug, slightly smaller in diameter than the inside diameter of the tubing, shall be inserted into a small section of tubing to spread it and form an approximate ellipse without stretching. The inside diameter and wall thickness of tubing shall then be determined by the use of a toolmaker's microscope.

NOTE.—If a toolmaker's microscope is not available at the manufacturer's plant for inspection tests, plug gages may be used for measurements of inside diameter of tubing. In the case of dispute, however, measurements with a toolmaker's microscope shall govern.

F–4b. *Dielectric break-down.*—

F–4b (1). *Test equipment.*—The test equipment shall consist of a well-designed, high-voltage transformer rated at not less than 2 kva. and the necessary auxiliary equipment for applying, controlling, and measuring the test voltage. The power supply shall consist of an alternating-current source having as nearly a true sine wave as possible at a frequency not exceeding 100 cycles per second.

F–4b (2). *Voltage control.*—The regulation shall be so controlled that the high-tension testing voltage taken from the secondary of the testing transformer can be raised gradually from any point and in no

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case more than 500 volts at a step. The control may be made by generator field regulation, with an induction regulator, or with a variable ratio auto-transformer. Any method of regulating the voltage is satisfactory which does not distort the wave more than 10 percent from a sinusoidal shape.

F–4b (3). *Voltage measurement.*—The voltage may be measured by any approved method which gives root-mean-square values, preferably by means of a voltmeter connected to a well designed tertiary coil in the testing transformer, or to a separate step-down instrument potential transformer. A voltmeter on the low tension side of the transformer is satisfactory if the ratio of the transformation does not change under any test condition. An electrostatic voltmeter properly calibrated in the high tension circuit is also satisfactory. A spark gap may be used to check the readings at very high potentials.

F–4b (4). *Test electrodes.*—

F–4b (4) a. *Forms F and S.*—The test electrodes for forms F and S materials shall consist of metal cylinders 2 inches in diameter and not less than ½ inch in length, with edges rounded to a radius of ¼ inch. The total weight on the sample shall be approximately 1 pound.

F–4b (4) b. *Form T.*—The test electrodes for form T material shall consist of bars 4 inches long and ¼ inch in width, rounded at the ends, and with all edges in contact with the sample rounded to a radius of $\frac{1}{32}$ inch. The total weight on the sample shall be approximately ½ pound.

F–4b (4) c. *Form U.*—The test electrodes for form U material shall consist of a solid conductor which will fit the inside of the tubing closely without stretching the tubing, and a tightly wrapped 1-inch strip of metal foil on the outside. In the case of flexible tubing of 0.315-inch inside diameter and larger, the tubing may be cut along its axis, flattened, and tested with the electrodes described in paragraph F–4b (4) b.

F–4b (5). *Specimens.*—The test specimens shall be 3½ inches square for forms F and S material, 6 inches long by the width of the material for form T, and 6 inches long for form U material. Tests shall be conducted at 25° C., immediately following removal from conditions C–96/25/0 and C–96/25/96.

F–4b (6). *Procedure.*—

F–4b (6) a. Tests shall be made with the specimen immersed in temperature-controlled transformer oil conforming to the requirements of Federal Specification VV–O–401. The transformer oil shall be changed frequently to prevent excessive carbonization. The test electrodes shall be kept clean and polished. They shall be self-aligning and shall be placed exactly opposite one another, with the specimen in a horizontal plane between them.

F–4b (6) b. Starting at zero, the voltage shall be increased uniformly to break-down, at a rate of 500 volts per second. The average dielectric break-down of the material shall be expressed in volts.

F–4c. *Dissipation factor and dielectric constant.*[6]—

F–4c (1). *Test equipment.*—The test equipment shall be suitable for measurement of dissipation factor and dielectric constant by the bridge

[6] For a complete discussion of theory, apparatus, electrodes procedure, and calculations, refer to ASTM–D–150–46T, Tentative Methods of Test for Power Factor and Dielectric Constant of Electrical Insulating Materials.

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method for frequencies of 10 cycles per second and by the resonant circuit method for frequencies of 10 cycles per second and higher. (Either method may be used at a frequency of 10 cycles per second.)

F–4c (1) a. *Bridge method.*—The bridge circuit used in this method shall be a type suitable for the measurements of dissipation factor and dielectric constant at frequencies up to 1 megacycle per second. Five bridge arrangements which have been found satisfactory for the purpose intended are the "Conjugate Schering Bridge," "High Voltage Schering Bridge," "Parallel Resistance Bridge," "Series Resistance Bridge," and "Transformer Bridge." It is not intended, however, that the use of other types of bridges of equal precision and convenience be excluded.

F–4c (1) b. *Resonant circuit method.*—The resonant circuit used in this method shall be a type suitable for the measurement of dissipation factor and dielectric constant at frequencies of 10 cycles per second and higher. Three resonant circuit arrangements which have been found satisfactory for the purpose intended are the "Susceptance Variation," "Resistance Variation," and "Resonant Rise." It is not intended, however, that the use of other types of resonant circuit arrangements of equal precision and convenience be excluded.

F–4c (1) c. *Sensitivity.*—The method used shall have an accuracy of determination of dielectric constant of ±5 percent and an accuracy of determination of dissipation factor of ±5 percent or 0.0002, whichever is greater.

F–4c (2). *Specimens.*—Where possible, specimens should be in the form of flat sheets. Larger sizes (1 inch or greater) of tubing may be slit along the axis of the tubing and flattened to obtain a sheet specimen. Suitable coaxial electrodes may be used on large tubing.

F–4d. *Volume resistivity.*[7]—

F–4d (1). *Apparatus.*—The apparatus shall consist of a source of direct-current potential of 180 volts, a galvanometer with suitable shunts having a minimum sensitivity of 0.0005 microamperes/millimeter at 1 meter-scale distance, a calibrating resistance, reversing switches, and keys. Any other apparatus may be used provided the same accuracy is obtained and the same voltage is applied.

F–4d (2). *Procedure.*—The electrodes shall be applied to the specimen and insulation resistance between guarded and unguarded electrodes measured by means of a test set which applies 180 volts direct current to the sample. The reading shall be taken after voltage has been applied for 1 minute.

F–4e. *Tensile strength, elongation, and percent return.*—

F–4e (1). *Tensile strength and elongation.*—

F–4e (1) a. *Apparatus.*—Tensile strength and elongation shall be determined on an approved pendulum-type testing machine. The machine shall be preferably power-driven.

F–4e (1) b. *Specimens.*—Ten samples of each size of material shall be tested after being subjected to conditions C–96/25/50 and E–400/100. Specimens cut from form F, and from forms S and T 0.010 inch and less in thickness, shall be 1 inch wide. Specimens cut from forms S and T greater than 0.010 inch in thickness shall be ½-inch dumbbell die-cut specimens, except that tape less than 1½ inches wide shall

[7] For a complete discussion of theory, apparatus specimens, electrodes, and calculation, refer to ASTM–D–257–46, Methods of Test for Insulation Resistance of Electrical Insulating Materials.

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be tested in full section. Form U tubing of 0.315-inch inside diameter and larger shall be tested using ½-inch dumbbell die-cut specimens.

F–4e (1) c. *Procedure.*—

F–4e (1) c (1). Bench marks 10 inches apart shall be used for materials having ultimate elongations of 25 percent or less. Bench marks 2 inches apart shall be used for materials having ultimate elongations greater than 25 percent.

F–4e (1) c (2). The test pieces shall be placed in the testing machine and stretched until the specimens break. The rate of separation of the jaws of the machine shall be 12±½ inch per minute. A scale shall be used to measure the elongation at instant of rupture. The tensile strength shall be reported in pounds per square inch.

F–4e (2). *Percent return.*—

F–4e (2) a. The tests shall be conducted on samples similar to those specified in paragraph F–4e (1) b, which have been subjected to condition C–96/25/50. The samples shall be stretched an amount equal to 75 percent of the elongation value after condition C–96/25/50, obtained when tested as described in paragraph F–4e (1) c (2). The stretch shall be maintained for a period of 100 seconds and the material allowed to recover for a period of 60 seconds. Three samples shall be tested.

F–4e (2) b. The rate of elongation and the distance between bench marks shall be as specified in paragraph F–4e (1) c.

F–4e (2) c. The percent return shall be expressed as the distance returned, multiplied by 100, and divided by the distance stretched.

F–4e (2) d. The percent set, expressed as the difference in length before and after stretching, multiplied by 100, and divided by the length before stretching, shall also be reported.

F–4f. *Bursting strength (type F, grades a and b, form U).*—

F–4f (1). *Apparatus.*—Bursting strength shall be measured on 9-inch lengths of tubing clamped over a source of compressed air at one end and plugged at the other end in such a manner that the clamps are 6 inches apart and that the tubing will not expand in a lengthwise direction.

F–4f (2). *Specimens.*—Three samples of each size of tubing, ½ to 1 hour after being subjected to condition D_s–48/50 (synthetic sea water), shall be subjected to each of the tests described in paragraph F–4f (3).

F–4f (2) a. *Synthetic sea water.*—Synthetic sea water used for this test shall be of the following composition. All chemicals used in its preparation shall be chemically pure (c. p.).

	Percent by weight
Sodium chloride (NaCl)	2.11
Magnesium chloride ($MgCl_2.6H_2O$)	0.558
Sodium bromide (NaBr)	.002
Potassium sulfate (K_2SO_4)	.065
Magnesium sulfate ($MgSO_4.7H_2O$)	.275
Calcium sulfate ($CaSO_4.2H_2O$)	.113
Distilled water	96.873

F–4f (3). *Procedure.*—

F–4f (3) a. *Increasing pressure.*—The pressure shall be increased in steps of 1 pound at the rate of 5 pounds per minute, until failure.

F–4f (3) b. *Continuous pressure.*—The pressure shall be increased in steps of 1 pound at the rate of 5 pounds per minute, until the desired

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continuous pressure shown in table XIV is attained. This pressure shall be maintained for 24 hours.

F–4g. *Effect of heat on material under tension (forms F, S, and T only).*—Brass tubing approximately 1 inch in diameter and 3¾ inches long shall be wrapped with two half-lapped windings of approximately 1-inch-width material under a tension of 500 pounds per square inch and the end secured. The specimens shall be subjected to condition E–400/100 and examined at intervals for cracking or splitting of layers.

F–4h. *Softening temperature (penetration).*—

F–4h (1). *Apparatus.*—The apparatus shall consist of an oven capable of raising the temperature of a stainless steel plate 4 inches long by 1¼ inches wide by ⅛ inch thick, at the rate of 1° C. every 2 minutes; a ⅛-inch magnetized steel rod, recessed at one end to hold a ⅛-inch ball bearing, in contact with and equipped to exert a pressure of 1,000 grams vertically down against the stainless steel plate, lying in a horizontal plane; an electrical circuit with a 110-volt source of electricity; a 110-volt glow lamp; and a suitable means of measuring the temperature of the steel plate as near the point of test application as is practicable. The use of a thermocouple to measure this temperature is suggested. A penetration tester as shown on figure 1 is recommended. The light C-clamp, containing the steel rod, counterbalance, and weight, shall be mounted on a bearing capable of giving the unit the necessary freedom of clockwise motion. With no load on the lower leg of the C-clamp, the counterbalance shall consist of a rider capable of being adjusted to neutralize the pressure of the ball bearing against the steel plate.

F–4h (2). *Specimens.*—Five 1-inch specimens shall be cut from the sample, and prepared for test. In the case of tubing, this may be accomplished by opening one side along the longitudinal axis.

F–4h (3). *Procedure.*—With no load on the rod, each specimen shall be inserted between the steel ball and steel plate, and in the case of tubing with the outside surface of the tubing facing the plate. The electrical circuit shall be connected in such a way that when the steel ball comes in contact with the plate, the lamp contained outside the oven shall light. The specimen in the oven at room temperature shall then be put under a compression load of 1,000 grams. The temperature of the steel plate shall be uniformly raised at the rate of 1° C. every 2 minutes until failure of the specimen, as indicated by the moment current flows through the circuit. The temperature at failure shall then be recorded as the softening temperature (penetration), to determine compliance with applicable requirements of table XIV.

F–4i. *Flexibility after aging.*—

F–4i (1). *Specimens.*—A sufficient number of 1-inch-width strips of forms F and S material and form T tape wider than 1 inch shall be used so that samples can be discarded after bending. Tape 1 inch wide and less, and form U tubing without conductor inside, shall be tested in full section.

F–4i (2). *Oven aging.*—The specimens shall be subjected to condition E–200/100 in a forced-draft oven, cooled to room temperature (25° C.), and bent as follows:

Forms F, S, and T—The material shall be bent 180° around a mandrel of a diameter equal to its own thickness.

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Form U—The tubing, in full section without conductor inside, shall be bent 180° around a ⅛-inch diameter mandrel.

F–4i (3). *Ultraviolet light aging (type F, grade b only).*—

F–4i (3) a. *Apparatus.*—The samples shall be secured to a light-colored corrosion-resistant metal disk approximately 17 inches in diameter and 0.1 inch in thickness, mounted on a phonograph turntable operating at 33 revolutions per minute. The disk shall be centered under an S–1 bulb which has been in use not less than 50 hours and not more than 550 hours, and which is mounted in a General Electric sun lamp, model BM12, or an equivalent lamp, equipped with a reflector approximately 15 inches in diameter at the lower rim. The lamp and turntable shall not be enclosed. The plane of the specimens shall be 6 inches from the bottom of the bulb.

F–4i (3) b. *Mounting of specimens.*—The specimens shall be supported about $\frac{3}{16}$ inch above the surface of the disk by a wire screen. Specimens shall be 4 inches long by not more than 1½ inches in width or diameter. The specimens shall be mounted radially with their inside end 3 inches from the center of the disk.

F–4i (3) c. *Procedure.*—After exposure to the ultraviolet radiation for 120 hours, the samples shall be removed and bent as described in paragraph F–4i (2).

F–4j. *Cold resistance.*—

F–4j (1). *Cold bend.*—

F–4j (1) a. *Specimens.*—One-inch-width strips of film and sheet, and tape wider than 1 inch, shall be used. Tape 1 inch wide and less, and tubing, shall be tested in full section.

F–4j (1) b. *Procedure.*—Specimens shall be conditioned for 1 hour in air, at the temperature specified below, and shall be bent while at that temperature as described in paragraph F–4i (2).

	°C.
Type F, grade b material	−40
All other types	−23

F–4j (2). *Cold vibration (type F, grade b, form U only).*—Type F, grade b tubing, which has been oven-aged as specified in paragraph F–4i (2), shall be tested for resistance to cracking during vibration at −55° C. The tubing shall be conditioned and brought to thermal equilibrium at −55° C., rigidly clamped and held stationary at one end, and rigidly clamped and vibrated at the other end. The distance between clamps shall be as determined by the laboratory. The tubing shall then be vibrated transversely for 15 minutes at a frequency of 60 cycles per second and an amplitude of 0.03 inch (0.06–inch excursion).

F–4k. *Lengthwise shrinkage.*—

F–4k (1). *Specimens.*—Three specimens of each size of material, strips ½ inch wide by 6 inches long with 5 inches between bench marks, shall be tested. Strips cut from forms F and S material shall be used. Tape and tubing shall be tested in full section.

F–4k (2). *Procedure.*—

F–4k (2) a. *Forms F, S, and T.*—The specimens shall be supported by a smooth-surfaced thin metal plate well covered with talc in an oven and subjected to condition E–2/100. The specimens shall then be removed and cooled to room temperature (25° C.). The length of the

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material before and after heating shall be measured and the percentage shrinkage calculated from these values.

F–4k (2) b. *Form U.*—The specimens shall be supported by insertion of a solid conductor having a diameter slightly less than the inside diameter of the tubing, and well coated with talc to prevent bonding. The specimens so supported shall be held horizontally in an oven and subjected to condition E–2/100. The specimens shall then be removed and cooled to room temperature (25° C.). The length of the material shall be measured before and after heating and the percent shrinkage calculated from these values.

F–4*l*. *Oil resistance—flexibility (not applicable to types A and D).*—

F–4*l* (1). *Specimens.*—Three specimens shall be tested.

F–4*l* (2). *Apparatus.*—Tests on material in forms F, S, T, and U shall be made in a closed container approximately 6 inches deep, filled with No. 3 high-swelling oil[8] to a line 2 inches below the cover. Strips of material in forms F, S, and T shall be hung in straight lengths with one end protruding through the cover and the lower end below the surface of the oil. Samples of material in form U shall be bent in the form of a U (around a diameter of approximately 4 inches), with ends protruding through the cover and the bottom of the U three inches below the surface of the oil. The sample then has a section immersed in oil, a section in oil vapor, and a section above the cover subjected to oven temperature only.

F–4*l* (3). *Procedure.*—The entire equipment shall be subjected to condition E–200/50. The samples shall then be removed and tested to determine compliance with the requirements of paragraph E–1d.

F–4m. *Corrosive effect.*—Forms F, S, and T, and form U tubing of 0.315-inch inside diameter and larger, shall be tested by butt-wrapping a ½-inch strip full length and return on a 6-inch piece of 0.162-inch (AWG No. 6) copper wire. Tubing of less than 0.315-inch inside diameter shall be tested by inserting the proper size wire into a 6-inch length of the tubing. Over the prepared specimens shall be wrapped a 30-inch length of 0.006-inch diameter bare copper wire in a bifilar winding. The ends of the wire shall be welded to heavy leads of 0.040-inch (AWG No. 18) copper wire and the sample placed in a test tube with the leads extending through a cork. A small amount of distilled water shall be placed in the test tube to maintain a high relative humidity and the cork shall be coated to produce a vaportight seal. The entire equipment shall be placed in an oven maintained at 70° C. The resistance of the 0.006-inch diameter copper wire shall be measured when the temperature of the sample reaches 70° C. and at 24-hour intervals for 720 hours. At the end of this period, samples shall be examined for visual evidence of corrosion on the 0.006-inch diameter wire. Percentage change in resistance versus time shall be plotted to show corrosion characteristics of the material.

F–4n. *Solvent and acid resistance (type F, grade b only).*—Type F, grade b material shall be dipped 10 times separately in each of the liquids specified in paragraph E–1f at 25° C. and allowed to dry for 30 minutes in free air at 25° C. after each dipping. The effect of the liquids on the material shall be determined by measurement of

[8] High-swelling oil is characterized by Saybolt universal viscosity of 155±5 seconds at 100° F., aniline point 157.1°±1.8° F. and flash point 330°±5° F. as described in ASTM D–471–46T, Tentative Methods of Test for Changes in Properties of Rubber and Rubber-Like Materials in Liquids.

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weight (where practicable) and of changes in dimensions, by visual examination of the plastic material, and by bending around a mandrel.

F–4o. *Chemical analysis.*—A chemical analysis shall be performed to identify the basic resin and the plasticizer, and their percentages in the formulation, as well as the percent of free carbon, chlorine, nitrogen, ash, and any element which may aid in characterizing the resin compound. Specific gravity, acetone, chloroform solubility, and refractive index shall also be determined.

F–4p. *Luminescence (type F, grade b only).*—

F–4p (1) . *Apparatus.*—An S–1 lamp shall be used as the ultraviolet source. The lamp shall be enclosed in a light-tight enclosure, the lower end of which is closed with an ultraviolet filter (Corning Glass No. 5860) to stop all stray light from affecting the determination. The filter shall be 1 inch from the lamp and the sample 3 inches from the lamp.

F–4p (2). *Procedure.*—Type F, grade b material shall be exposed for 30 minutes to the ultraviolet light from the S–1 lamp. Fluorescence shall be determined during the 30-minute exposure period. Phosphorescence shall be determined immediately after removal of the light at the end of the 30-minute exposure period.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packaging.*—Unless otherwise specified, the subject commodity shall be packaged as follows:

G–1a. *Form F.*—Film in rolls shall be individually wrapped in not less than 30-pound basis weight kraft paper, or equal, waxed on one side only (waxed side out) and securely sealed with 2-inch-width, water-resistant tape conforming to the requirements of Joint Army-Navy Specification JAN–P–127, type III, grade B. Rolls shall be further packaged in a suitable style 200-pound Mullen test corrugated "A" flute or solid fiberboard box conforming to the requirements of Joint Army-Navy Specification JAN–P–108, with adequate corrugated pads of the same material protecting the ends of the rolls against damage in shipment and storage.

G–1b. *Form T.*—Tape wound on cores shall be separated by chipboard disks, or equal, to prevent dishing or telescoping of rolls. Rolls shall be further wrapped in bundles with not less than 30-pound basis weight kraft paper, or equal, waxed on one side only (wax side out) and securely sealed with 2-inch-width, water-resistant tape conforming to the requirements of Joint Army-Navy Specification JAN–P–127, type III, grade B. The length of the bundle shall not exceed 24 inches. Bundles shall be further packaged in a suitable style 200-pound Mullen test corrugated "A" flute or solid fiberboard box conforming to the requirements of Joint Army-Navy Specification JAN–P–108.

G–1c. *Form S.*—Sheets shall be packaged flat and adequately protected against damage in shipment and storage by use of chipboard separators or other cushioning material and wrapped in 30-pound basis weight kraft paper, or equal, waxed on one side only (waxed side out) and securely sealed with 2-inch-width water-resistant tape conforming to the requirements of Joint-Army-Navy Specification JAN–P–127, type III, grade B. When furnished in rolls, they shall be packaged in conformance with the requirements of paragraph G–1a or G–1b, as applicable.

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G–1d. *Form U.*—Tubing, in bundles of straight lengths or on reels, shall be individually wrapped in not less than 30-pound basis weight kraft paper, or equal, waxed on one side only (waxed side out), and securely sealed with 2-inch-width, water-resistant tape conforming to the requirements of Joint Army-Navy Specification JAN–P–127, type III, grade B. Tubing may be furnished in coils, securely tied to prevent unwinding; no wrapping will be required. Tubing shall be further packaged in suitable style corrugated fiberboard boxes conforming to the requirements of Joint Army-Navy Specification JAN–P–108, having a minimum Mullen test of 200 pounds.

G–2. *Packing.*—Where practicable, shipping containers shall be of uniform size, and shall contain an identical number of items of one form, size, type, and color. Containers shall be designed so that the contents fit in a compact manner.

G–2a. *For domestic shipment.*—Unless otherwise specified, the subject commodity packaged as described in paragraph G–1, shall be packed in fiberboard boxes, cleated plywood boxes, or nailed wood boxes, or wire-bound wood boxes,[9] as follows:

G–2a (1). Corrugated "A" flute or solid fiberboard boxes shall conform to the requirements shown in table XII.

TABLE XII.—*Requirements for fiberboard boxes.*

Gross weight (maximum)	Average bursting strength (minimum)	Sum of inside dimensions (maximum)
Pounds	*Pounds*	*Inches*
40	200	75
65	275	90
90	350	100

All flaps shall be sealed by means of a suitable adhesive, or shall be metal-stitched or taped. If taped, water-resistant tape 2½ inches wide conforming to the requirements of Joint Army-Navy Specification JAN–P–127, type III, grade B, shall be used.

G–2a (2). Cleated plywood boxes, nailed wood boxes, and wirebound wood boxes [9] shall conform to the requirements of Joint Army-Navy Specifications JAN–P–105, JAN–P–106, and JAN–B–107, respectively. Each box shall be strapped in conformance with requirements of the appendix of the applicable box specification. The gross weight shall not exceed approximately 150 pounds.

G–2a (3). Form F material, wrapped in 30-pound basis weight kraft paper, waxed on one side only, may be packed in a suitable cylindrical fiberboard container, securely capped and sealed. The contents shall be adequately protected against damage in shipment and storage by the use of fiberboard pads at the ends of the roll.

G–2b. *For overseas shipment.*—Unless otherwise specified, the subject commodity, packaged as described in paragraph G–1, shall be packed in cleated plywood boxes,[10] nailed wood boxes, wirebound wood

[9] Not applicable to Air Force purchases.
[10] For Air Force shipments, plywood shall be type B, condition I, of Joint Army-Navy Specification JAN–P–139.

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boxes,[11] or fiberboard boxes,[11] conforming to the requirements of Joint Army-Navy Specifications JAN–P–105, JAN–P–106, JAN–B–107, and JAN–P–108, respectively. The gross weight of wood boxes shall not exceed approximately 150 pounds; of fiberboard boxes, approximately 40 pounds. Each box shall be lined with a sealed waterproof bag, or its equivalent, made from material conforming to the requirements of Joint Army-Navy Specification JAN–P–125 for case liners. The seams and closures shall have water-resistance equal to that of the body material. Each box shall be strapped in accordance with the requirements of the appendix of the applicable box specification.

G–3. *Marking.*—

G–3a. *Packages.*—

G–3a (1). *Army and Air Force.*—Unless otherwise specified, all Army purchases shall be marked as specified by the procuring service; for Air Force shipments, marking of interior packages shall be in accordance with U. S. Air Force Specification 40985.

G–3a (2). *Navy.*—Unless otherwise specified, packages shall be marked in accordance with the requirements of the Navy Shipment Marking Handbook.

G–3b. *Shipping containers.*—In addition to any special marking required by the contract or order, shipments for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2, for the Air Force in accordance with the requirements of U. S. Army Specification 94–40645, and for the Navy in accordance with the requirements of the Navy Shipment Marking Handbook.

H. NOTES

H–1. *Intended use.*—

H–1a. *Type A.*—This type of material is intended for ultrahigh-frequency applications. It possesses excellent electrical properties which remain practically constant over a wide range of frequencies, temperatures, and humidities, and a very low loss factor.

H–1b. *Type B.*—This type of material is intended for applications requiring a good dielectric strength, fairly low loss factor, relatively low percent moisture absorption, and good insulation resistance.

H–1c. *Type C.*—This type of material is intended for applications requiring good dielectric strength, fairly low loss factor, and a relatively high degree of heat resistance with some sacrifice in moisture resistance.

H–1d. *Type D.*—This type of material is intended for radio-frequency applications requiring very good dry dielectric strength, fairly low loss factor, and fair moisture resistance.

H–1e. *Type E.*—This type of material is intended for applications having excellent mechanical strength and excellent bursting strength. It is intended for general purpose electrical applications but is more rigid than type F.

H–1f. *Type F, grade a.*—This type of material is intended as a high-elongation, elastic material with good electrical properties, suitable as an alternate for rubber insulating tape in cable or other applications.

H–1g. *Type F, grade b.*—This type of material is intended as a high-elongation, elastic material similar to type F, grade a, with less ri-

[11] Not applicable to Air Force purchases.

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ted States Army, Joint Army-Navy, and Federal Specifications and ndards." Copies of this Index may be obtained from the Super- ndent of Documents, Government Printing Office, Washington 25, C. Services within the Department of the Army and the Depart- nt of the Air Force will obtain copies of National Military Estab- ment, Joint Army-Navy, U. S. Army, and Federal specifications ough established departmental channels. Both the title and identi- ng symbol number should be stipulated when requesting copies. [–10. Copies of National Military Establishment, Joint Army-Navy, . Federal specifications (required for Navy purchases), Navy De- tment specifications, and the Navy Shipment Marking Handbook, y be obtained upon application to the Bureau of Supplies and ounts, Navy Department, Washington 25, D. C., except that Naval vities should make application to the Supply Officer in Command, al Supply Center, Norfolk 11, Va. Both the title and identifying bol number should be stipulated when requesting copies.

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TABLE XIV.—*Property values of synthetic-resin electrical insulation.*

Property to be tested	Test procedure paragraph	Condition prior to test	Unit	Nominal size of material (inch)	Type A	Type B	Type C	Type D	Type E	Type F	
										Grade a	Grade b
Dielectric break-down (NOTE.—For thickness not shown, the dielectric break-down value shall be that given for the nearest thickness shown.)	F-4b	C-96/25/0	Kilovolts (min.).	Thickness:							
				0.0005	4.5	1.5	1.5	2.0	0.90	0.60	0.60
				0.001	6.0	2.7	2.7	3.4	1.6	1.0	1.0
				0.0015	7.4	3.5	3.5	5.0	2.1	1.4	1.4
				0.002	8.3	4.3	4.3	6.8	2.6	1.8	1.8
				0.003	10.0	5.6	5.6	8.6	3.6	2.6	2.6
				0.004	11.0	6.8	6.8	10.1	4.5	3.5	3.5
				0.005	12.0	7.8	7.8	11.4	5.4	4.3	4.3
				0.008	14.7	10.4	10.4	14.2	7.8	6.0	6.0
				0.010	16.3	11.8	11.8	15.5	9.4	7.6	7.6
				0.015	19.0	15.1	15.1	18.5	12.9	10.5	10.5
				0.020	22.5	17.9	17.9	20.9	15.9	13.6	13.6
				0.025	24.0	20.4	20.4	23.0	18.6	15.6	15.6
				0.030	26.3	22.4	22.4	24.8	20.8	17.6	17.6
				0.040	30.0	25.6	25.6	28.0	24.0	20.6	20.6
				0.050	33.0	27.8	27.8	30.7	25.7	22.8	22.8
				0.060	35.8	29.3	29.3	32.5	27.0	24.0	24.0
		C-96/25/96			(1)	(1)	(2)	(3)	(4)	(5)	(6)
Dissipation factor at 1 kc. (forms F, S, and U).	F-4c	C-96/25/0	Maximum		0.0006	0.02	0.035	0.015	0.08	0.14	0.20
		C-96/25/96			0.0008	0.03	0.045	0.03	0.08	0.16	0.20
Dissipation factor at 1 mc. (forms F, S, and U).	F-4c	C-96/25/0	Maximum		0.0006	0.045	0.05	0.028			
		C-96/25/96			0.0008	0.055	0.07	0.045			
Dielectric constant at 1 kc. (forms F, S, and U).	F-4c	C-96/25/0	Maximum		2.5	4.0	4.5	4.0	5.0	7.0	7.0
		C-96/25/96			2.7	5.0	8.0	5.0	6.0	8.0	8.0
Dielectric constant at 1 mc. (forms F, S, and U).	F-4c	C-96/25/0	Maximum		2.5	4.0	4.3	4.0			
		C-96/25/96			2.7	5.0	7.0	5.0			
Volume resistivity (forms F, S, and U).	F-4d	C-96/25/0	Ohm-centimeters.		10^{17}	10^{13}	10^{13}	10^{14}	10^{13}	10^{11}	10^{11}
		C-96/25/96			10^{16}	10^{12}	10^{12}	10^{12}	10^{12}	10^{10}	10^{10}
Tensile strength	F-4e (1)	C-96/25/50	Hundreds of pounds per square inch (min.).		14	50	60	35	35	18	18
Elongation	F-4e (1)	C-96/25/50	Percent (min.)		3.5×10^2	25	10	25	15	2.0×10^2	2.0×10^2
		C-96/25/50	Percent (max.)			1.0×10^2	50	1.0×10^2	1.0×10^2	4.5×10^2	4.5×10^2
		E-400/100	Percent change (max.).							35	45
Return	F-4e (2)	C-96/25/50	Percent (min.)	Forms F, S, and T all thicknesses.						75	65
				Form U wall thickness:							
				0.020						60	50
				0.030						50	40
				0.040 and up.						45	35
Bursting strength (form U for cable sizes 3,000–800,000 cm. inclusive). Increasing pressure.	F-4f	Ds-48/50 (synthetic sea water)	Pounds per square inch (min.)	Inside diameter (min.):							
				0.155						96	96
				0.245						59	59
				0.275						53	53
				0.315						46	46
				0.355						41	41
				0.395						58	58
				0.425						54	54
				0.445						50	50
				0.515						44	44
				0.565						54	54
				0.625						49	49
				0.675						45	45
				0.750						41	41
				0.810						47	47
				0.880						43	43
				0.950						40	40
				1.020						38	38
				1.090						35	35
				1.155						40	40
				1.205						38	38
				1.260						36	36
				1.345						34	34
				1.455						32	32
				1.565						29	29
Continuous pressure										(7)	(7)
Softening temperature (penetration).	F-4h	C-96/25/50	Degrees C. (min.).		65	120	160	105	120	75	65
Lengthwise shrinkage	F-4k	E-2/100	Percent (max.)		2.0	3.0	3.0	2.0	5.0	10.0	10.0

[1] 85 percent of dry value specified.
[2] 75 percent of dry value specified.
[3] 70 percent of dry value specified.
[4] 95 percent of dry value specified.
[5] 90 percent of dry value specified.
[6] 80 percent of dry value specified.
[7] One-half of above value.

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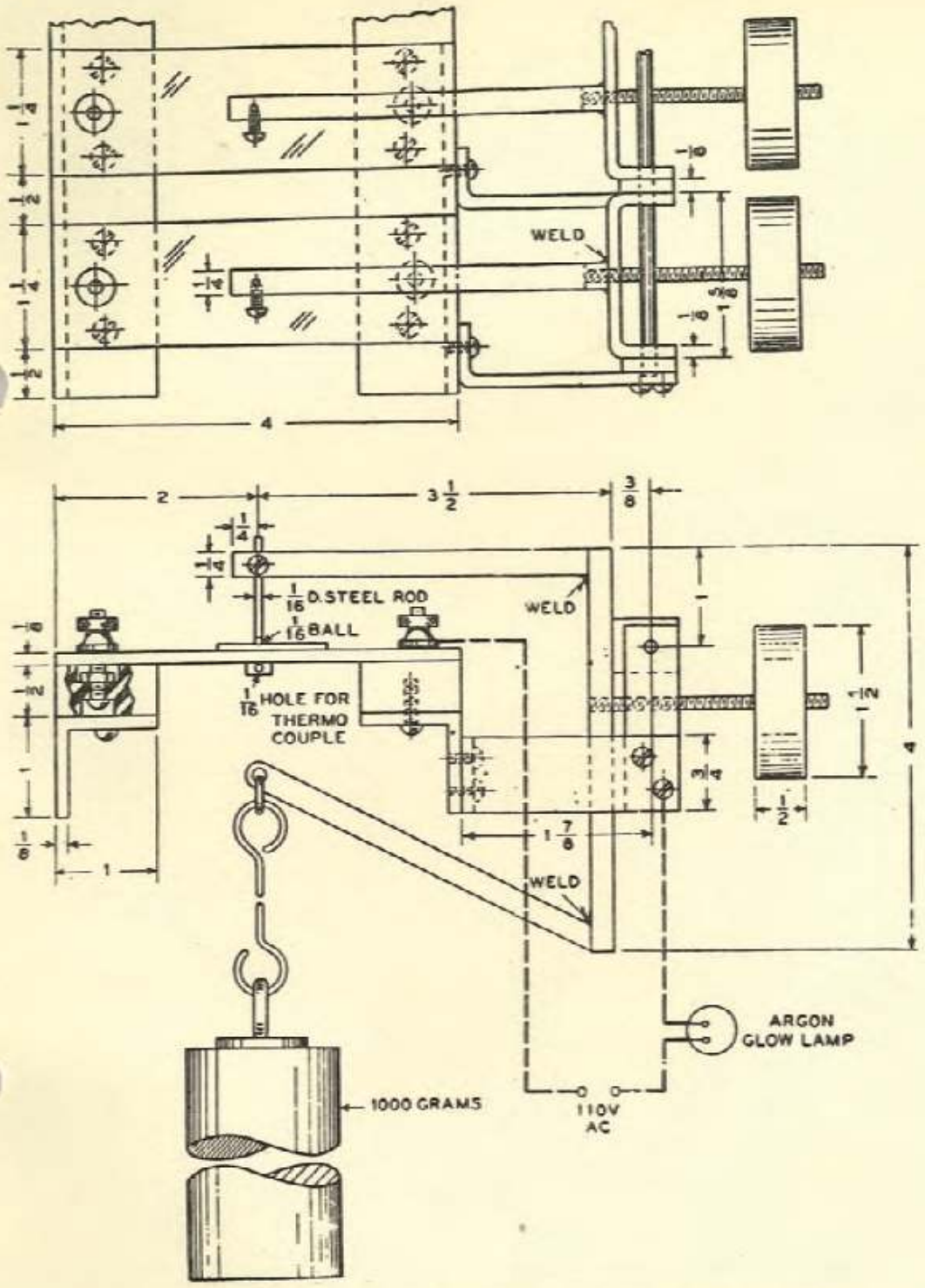


FIGURE 1.—Equipment for softening temperature (penetration) test.

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